

N8N via DOCKER LEARNING ROADMAP

SECTION 1 — FOUNDATIONAL LEARNING

Learning Goals:

- Understand what **n8n** is: an open-source, self-hostable automation platform similar to Zapier.
- Learn workflows, nodes, triggers, and actions.
- Grasp the difference between static automations and AI-powered logic flows.

Action Items:

1. **Install & Deploy:** Install globally via `npm install -g n8n` or run temporarily with `npm run n8n`. Alternatively, use Docker for a stable setup (`docker run -it --rm -p 5678:5678 n8nio/n8n`).
2. **Explore Templates:** Import and customize built-in workflow templates. Rename nodes for clarity and readability.
3. **Create a Starter Workflow:** Build a "Daily Weather Email" automation using a Schedule Trigger, OpenWeather API, and Gmail node.

SECTION 2 — INTERMEDIATE AUTOMATIONS

Learning Goals:

- Understand node types and how data flows between input → processing → output.
- Manipulate JSON data within nodes.
- Use logic and branching for smart, dynamic routing.

Action Items:

1. **Logic Control:** Build a form-driven routing system using a **Switch** node to route by conditions like budget or priority.
2. **Data Storage:** Connect to **Google Sheets** and append rows automatically when data is received.
3. **Data Hygiene:** Use a **Set** node to clean or reformat user input. Apply JSON expressions like `{{ $json["field"] }}`.

4. **Debugging Practice:** Use **Pin Data**, **Table**, **Schema**, and **JSON** views to inspect and troubleshoot workflow data.
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SECTION 3 — ADVANCED INTELLIGENT WORKFLOWS

Learning Goals:

- Simulate AI agent behavior using LLMs (e.g., GPT-4) and connected tools.
- Understand workflow modularity and sub-workflow connections.
- Integrate APIs for dynamic, data-driven tasks.

Action Items:

1. **Build an AI Personal Assistant:** Combine **OpenAI**, **Calendar**, **Gmail**, and **Sheets** nodes for context-aware task handling.
 2. **Sub-Workflow Modularization:** Create independent workflows triggered by others using the **Call Workflow** node.
 3. **Custom API Integrations:** Add **HTTP Request** nodes to connect custom APIs or external services.
 4. **Testing & Scaling:** Chain multi-step logic flows, monitor execution logs, and refine for performance.
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SECTION 4 — CONTINUOUS IMPROVEMENT & INNOVATION

Learning Goals:

- Build habits around clean, modular, and maintainable workflow design.
- Learn how to integrate retrieval-augmented generation (RAG) using external vector databases.

Action Items:

1. **Iterate Daily:** Keep a sandbox workflow for experimenting with new nodes or ideas.
2. **Use ChatGPT or LLM Tools:** Co-develop, debug, and optimize workflow logic.
3. **Explore Advanced Topics:** Experiment with RAG setups using APIs like **Pinecone**, **Weaviate**, or **Supabase Vector**.
4. **Document Everything:** Maintain a workflow log capturing purpose, inputs, outputs, and troubleshooting notes.

SECTION 5 — ENVIRONMENT SETUP & MAINTENANCE

Learning Goals:

- Deploy and maintain n8n locally using Docker.
- Understand Docker images, containers, ports, and environment variables.

Action Items:

1. **Prepare Environment:** Install Docker Desktop and verify with `docker --version`.
2. **Pull n8n Image:** In Docker Desktop, search for `n8nio/n8n` and click **Pull**.
3. **Run Container:** Name it `n8n_local`, expose port `5678`, and start the container.
4. **Access n8n:** Visit `http://localhost:5678` to set up your admin account.
5. **Secure Configuration:** Use `.env` files for API keys and credentials—never hardcode secrets in nodes.
6. **Maintain Environment:** Use Docker commands to stop, start, and update containers as needed.
7. **Mini Projects:** Recreate the weather workflow, test webhooks, or deploy an AI assistant locally.

Prerequisites:

- Familiarity with JSON, APIs, and basic JavaScript expressions.
- GitHub account for version-controlled workflow exports.

End Goal:

Achieve proficiency in designing, automating, and scaling n8n workflows across both traditional automation and AI-enhanced use cases.

PLATFORM NOTES — MAC & WINDOWS COMPATIBILITY

macOS:

- Install Node.js via Homebrew, MacPorts, or directly from nodejs.org.
- Use Terminal for commands; n8n, npm, and Docker all run natively on Apple Silicon and Intel Macs.

- Store environment variables in `~/.bash_profile`, `~/.zshrc`, or a `.env` file.
- Access the UI through `http://localhost:5678` in any browser.

Windows:

- Install Node.js from nodejs.org (includes npm by default).
- Use Docker Desktop with **WSL2** enabled; ensure virtualization is active in BIOS.
- Commands work in PowerShell or Windows Terminal, but environment variable syntax differs:

macOS/Linux → `export API_KEY=yourkey`

Windows (PowerShell) → `$env:API_KEY = "yourkey"`

- Access n8n via `http://localhost:5678` just as on macOS.

Both platforms support identical n8n functionality. The only practical differences are installation steps and environment variable handling.